FAT LAB AGILE DEVELOPMENT PROCESS AND DEVOPS

FAT

SET-1

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1.Develop order and processing logic using java

2.write Junit test for order success and failure

3.Implement CI/CD pipeline with automated testing

- Setup version control using git
- Configure CI pipeline
- Automate builds using maven
- Integrate automated tests in pipeline
- Configure CD pipeline for automated pipeline for automatic deployment to staging /production

4.Deploy services on Docker and kubernetes

- Containerize application using Docker
- Build and push docker image to container repositry
- Create kebernetes deployment and yaml file

ANSWER :

Code Logic:

Agile Fat2 — OrderProcessor.java

src > main > java > com > agilefat2 > OrderProcessor.java > OrderProcessor > processOrder(String, int, double)

```java
package com.agilefat2;

import java.util.HashMap;
import java.util.Map;
public class OrderProcessor {
    private final Map<String, Double> inventory;
    private final double paymentThreshold = 0.01;

    public OrderProcessor() {
        this.inventory = new HashMap<>();
        inventory.put("item1", 10.0);
        inventory.put("item2", 5.0);
        inventory.put("item3", 0.0);
    }

    /**
     * Processes an order.
     * @param itemId the item to order
     * @param quantity the quantity
     * @param paymentAmount the payment amount
     * @return true if order is successful, false otherwise
     */
    public boolean processOrder(String itemId, int quantity, double paymentAmount) {
        if (itemId == null || itemId.trim().isEmpty()) {
            return false;
        }
        if (quantity <= 0) {
            return false;
        }
        if (paymentAmount < paymentThreshold) {
            return false;
        }

        Double available = inventory.get(itemId);
        if (available == null || available < quantity) {
            return false;
        }
        inventory.put(itemId, available - quantity);
        return true;
    }
```

```
gowreesha@Gowreeshs-MacBook-Air Agile Fat2 % echo "# FatLab" >> README.md …
branch 'main' set up to track 'origin/main'.
gowreesha@Gowreeshs-MacBook-Air Agile Fat2 %
```

Junit Test for success and failure :

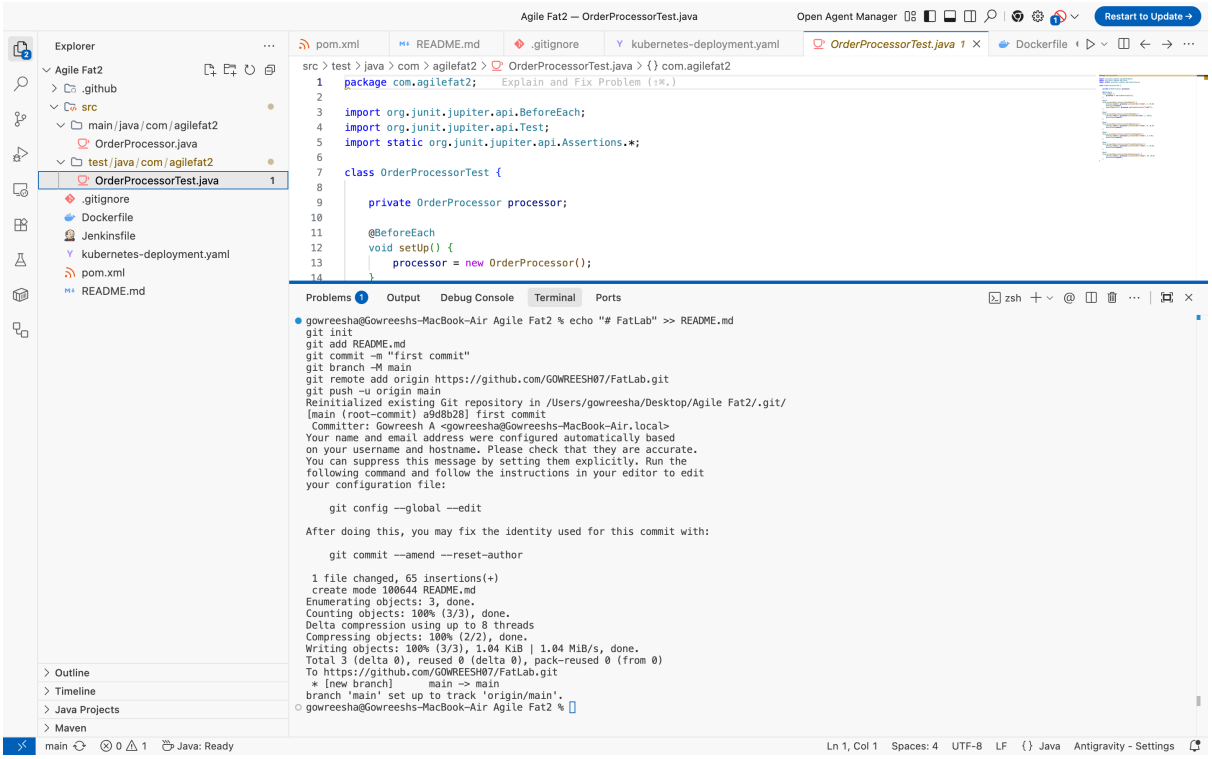
Agile Fat2 — OrderProcessorTest.java

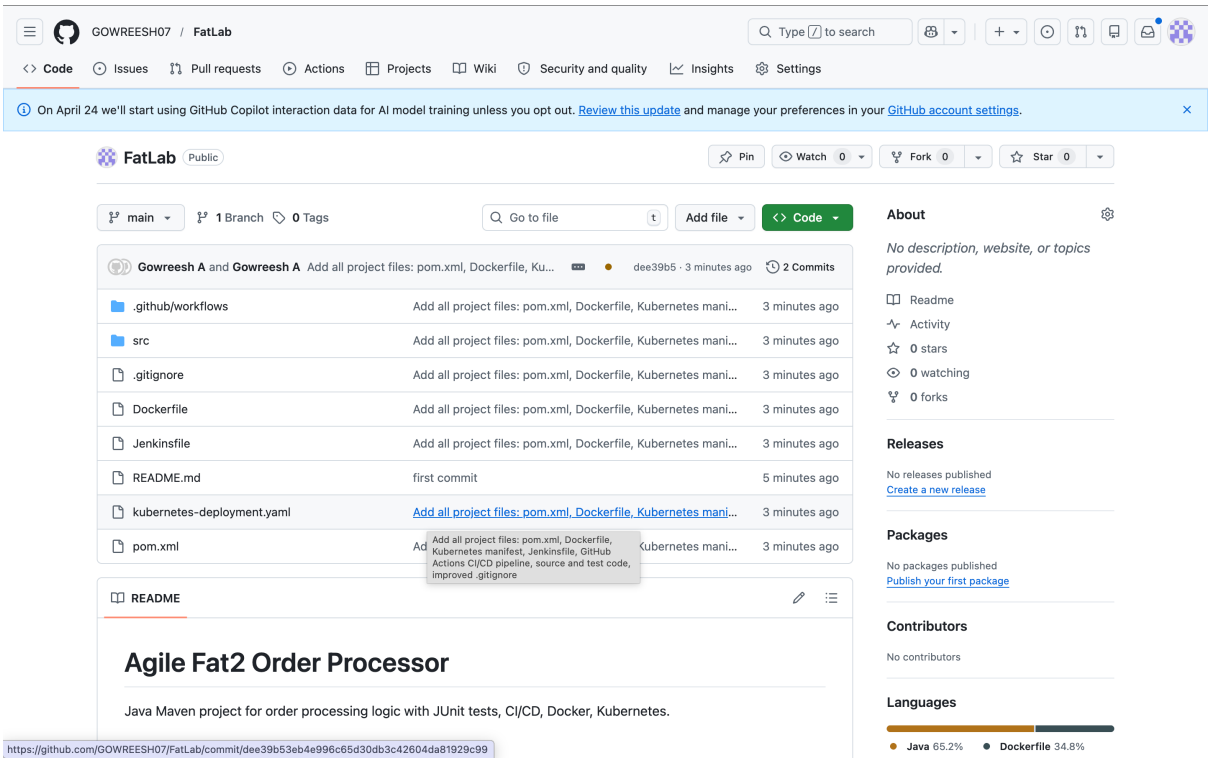
src > test > java > com > agilefat2 > OrderProcessorTest.java > {} com.agilefat2

```java
package com.agilefat2;

import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

class OrderProcessorTest {

    private OrderProcessor processor;

    @BeforeEach
    void setUp() {
        processor = new OrderProcessor();
    }

    @Test
    void processOrder_Success_ValidInputs() {
        boolean result = processor.processOrder("item1", 2, 10.0);
        assertTrue(result);
        assertEquals(8.0, processor.getInventoryLevel("item1"));
    }

    @Test
    void processOrder_Failure_InvalidItemId() {
        boolean result = processor.processOrder(null, 1, 10.0);
        assertFalse(result);
    }

    @Test
    void processOrder_Failure_InvalidQuantity() {
        boolean result = processor.processOrder("item1", 0, 10.0);
        assertFalse(result);
    }

    @Test
    void processOrder_Failure_InsufficientPayment() {
        boolean result = processor.processOrder("item1", 1, 0.0);
        assertFalse(result);
    }
```

```
gowreesha@Gowreeshs-MacBook-Air Agile Fat2 % echo "# FatLab" >> README.md …
branch 'main' set up to track 'origin/main'.
gowreesha@Gowreeshs-MacBook-Air Agile Fat2 %
```

Github Pushing Screenshot:





CI/CD pipeline:

Jenkins / FAt / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

try sample Pipeline...

```
12
13    environment {
14
15        GITHUB_REPO        = 'https://github.com/GOWREESH89/FatLab'  // ← update
16
17
18        DOCKER_HUB_USER    = credentials('DOCKERHUB_CREDENTIALS').username
19        DOCKER_IMAGE       = "${DOCKER_HUB_USER}/agile-fat2"
20
21
22        K8S_NAMESPACE_STG  = 'staging'
23        K8S_NAMESPACE_PROD = 'production'
24        K8S_DEPLOYMENT     = 'order-processor'
25
26
```

Use Groovy Sandbox ?

Pipeline Syntax

Advanced

Advanced

Save Apply

Jenkins / medicine-inventory / #18

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#18'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

#18 (Apr 12, 2026, 10:39:23 PM)

Add description | Keep this build forever

Started by user ABIRAMI

Started 3 min 53 sec ago
Took 1 min 13 sec

This run spent:

- 23 ms waiting;
- 1 min 13 sec build duration;
- 1 min 13 sec total from scheduled to completion.

git

Revision: 658f5312a66f515009f61ba2ba765c62c4a25e54
Repository: https://github.com/abiramis2023-ux/medicine-inventory.git

- refs/remotes/origin/main

No changes.

#18

```
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\medicine-inventory
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clean Workspace)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] deleteDir
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
```

#18

```
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\medicine-inventory>mvn -version
Apache Maven 3.9.14 (996c630dbc656c76214ce58821dcc58be960875b)
Maven home: C:\ProgramData\Jenkins\.jenkins\tools\hudson.tasks.Maven_MavenInstallation\Maven
Java version: 17.0.12, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-17
Default locale: en_IN, platform encoding: Cp1252
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\medicine-inventory>mvn clean package -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.inventory:medicine-inventory >------------------
[INFO] Building medicine-inventory 1.0
[INFO]   from pom.xml
```

#18

```
6e8d4ea1b4bf: Pushed
6b48f7e4777d: Pushed
2edb5c1a24b0: Pushed
6828893045b8: Pushed
bf574942e531: Pushed
de47083ed7d7: Pushed
94c27faa059b: Pushed
latest: digest: sha256:1e32470d1052195b320bdccfb3f890c1dfc99e517ceca89d44418ec045bbbc0f size: 856
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
 CI/CD Pipeline SUCCESS!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Maven build process:

Github Link:

https://github.com/GOWREESH07/FatLab

Docker File:

Kubernetes Deployment and YAML files:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: order-processor
  labels:
    app: order-processor
    version: "1.0.0"
spec:
  replicas: 2
  selector:
    matchLabels:
      app: order-processor
```

```yaml
  template:
    metadata:
      labels:
        app: order-processor
    spec:
      containers:
        - name: order-processor
          image: your-dockerhub-username/agile-fat2:latest   # ← replace with your
Docker Hub username
          imagePullPolicy: Always
          ports:
            - containerPort: 8080
          resources:
            requests:
              memory: "256Mi"
              cpu: "250m"
            limits:
              memory: "512Mi"
              cpu: "500m"
          readinessProbe:
            httpGet:
              path: /health
              port: 8080
            initialDelaySeconds: 10
            periodSeconds: 5
          livenessProbe:
            httpGet:
              path: /health
              port: 8080
            initialDelaySeconds: 30
            periodSeconds: 10
apiVersion: v1
kind: Service
metadata:
  name: order-processor-service
  labels:
    app: order-processor
spec:
  type: LoadBalancer
  selector:
    app: order-processor
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: order-processor-hpa
spec:
  scaleTargetRef:
```

```yaml
    apiVersion: apps/v1
    kind: Deployment
    name: order-processor
  minReplicas: 2
  maxReplicas: 5
  metrics:
    - type: Resource
      resource:
        name: cpu
        target:
          type: Utilization
          averageUtilization: 70
```

Docker Screenshots:

```
dicine-inventory .
[+] Building 7.2s (9/9) FINISH
 => [internal] load bui  0.0s
 => => transferrin 195B  0.0s
 => [internal] load met  1.7s
 => [auth] library/ecli  0.0s
 => [internal] load .do  0.0s
 => => transferring  2B  0.0s
 => [1/3] FROM docker.i  0.0s
 => => resolve docker.i  0.0s
 => [internal] load bui  2.7s
 => CACHED [2/3] WORKDI  0.0s
 => [3/3] COPY target/m  0.1s
e system: exec: "git.exe": executable file not found in %PATH%
```

```
Using default tag: latest
4425c0a98b6a: Pushed
de47083ed7d7: Layer already exbf574942e531: Layer already exists ea1b4bf: Layer already ex56e1dc442689: Layer
already ex52a553dd3db9: Pushed          4425c0a98b6a: Pushing  25.17MB/47.7MBa059b: Layer already ex6828893045
latest: digest: sha256:3cf840ecb45de9cd3fbb7bdb52ee719e9acec9f2607fb40e96e6764b4c904741 size: 856
PS C:\Users\wellw\OneDrive\Documents\Desktop\WINTER-(25-26)\Agile\gowreesh_agile>
```

ktop-cloud-provider-kind	v0.5.0	4ad59ce20658
/envoy	v1.36.4	08acc53db6f4
ktop-containerd-registry-mirror	v0.0.3	179a9ea3ae47
/medicine-inventory	latest	1e32470d1052

Kubernetes:

Deployments

Name	Status	Pods
reminder-v1	Available	1/1
reminder-v2	Available	1/1
reminder-v3	Available	1/1
springboot-app	Available	1/1
medicine-app	Available	2/2

Pods

Name	Status
reminder-v2-5781789765-ks6j4	Running
reminder-v3-74dc7f5747-7cjzt	Running
springboot-app-6b8d8bb47b-bvs9z	Running
medicine-app-79c67c7d5d-sml94	Running
medicine-app-79c67c7d5d-lmbxw	Running